v1.0Gluon Deep Dive Emulation Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Gluon Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: `v1.0Gluon Deep Dive Emulation Request V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

---

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Gluon spin fraction	Delta_g ~ 0.2 (V1.0)	STAR 2024: Delta_g(0.05<x<0.2) = +0.199±0.046 confirmed. Plus OAM contribution L_g ~ 0.2*hbar	MEDIUM	OAM component requires new data field in gluon state
Gluon saturation (CGC)	Asymptotic freedom only	LHCb 2024 forward region shows saturation. CGC mode triggered when gluon density > Q_s. New condensate state	HIGH	Missing entire physics regime: gluon saturation changes simulation behavior at high density
Glueball candidates	Not in V1.0	BESIII 2023: X(2370) pseudoscalar glueball candidate. LQCD: lightest glueball 1710 MeV. GLUEBALL particle type added	HIGH	Entirely new particle class — purely gluonic — absent from V1.0

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Triple gluon vertex visual	Conceptual	V2.0: g→gg splitting implemented as Y-shaped worldline. Branching when E > 2*Lambda_QCD	MEDIUM	Visual representation of core QCD process missing from V1.0

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Glueball, Gluon
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

---

Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Gluon Deep Dive Emulation Request V1.0 archived | V2.0 is the active specification as of June 2026